

$$x = \frac{dx}{dt} = v$$

$$v = \frac{dv}{dt} = a$$

$$\Sigma f = ma$$

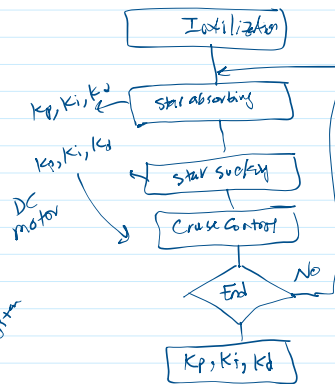
$$u - b\dot{v} = m\ddot{v}$$

$$\ddot{v} = \frac{1}{m}u - \frac{b}{m}\dot{v} \quad \text{--- (1)}$$

let $y_1 = x$
 $y_2 = \dot{y}_1 = \dot{x} = v$
 $\dot{y}_2 = \ddot{v} = \frac{1}{m}u - \frac{b}{m}y_2$

$$\begin{cases} \dot{y}_1 = y_2 \\ \dot{y}_2 = \frac{1}{m}u - \frac{b}{m}y_2 \end{cases}$$

this is the system



```
function ym = CruiseControl_lead(t, x)
% Uncontrolled Model
m = 1000;
b = 50;
V_ref = 10;
u = V_ref;
ym = zeros(2,1);
y(1) = x(1);
y(2) = b/m*x(1)+1/m*u;
end
```



$$e = V_{ref} - y$$

$$e = V_{ref} - y_2$$

$$u = K_p e \rightarrow \text{Proportional control}$$

P-Controller :-



PI-Controller :-

$$e \rightarrow \left[K_p + \frac{K_i}{s} \right] \rightarrow u = K_p e + \frac{K_i}{s} e$$

$$u = K_p e + K_i \int e dt$$

$$\begin{cases} \dot{y}_1 = y_2 \\ \dot{y}_2 = \frac{1}{m}u - \frac{b}{m}y_2 \\ \dot{y}_3 = V_{ref} - y_2 \\ u = K_p (V_{ref} - y_2) + K_i y_3 \end{cases}$$

$$\begin{cases} \text{let } y_3 = \int e dt \\ \dot{y}_3 = e \\ e = V_{ref} - y_2 \end{cases}$$

PID-Controller :-

$$e \rightarrow \left[K_p + \frac{K_i}{s} + sK_d \right] \rightarrow u$$

$$u = K_p e + K_i \int e dt + K_d \frac{de}{dt}$$

$$\begin{cases} \dot{y}_1 = y_2 \\ \dot{y}_2 = \frac{1}{m}u - \frac{b}{m}y_2 \\ \dot{y}_3 = e \\ \dot{y}_4 = -\dot{y}_3 \end{cases}$$

$$\begin{cases} \text{let } y_3 = \int e dt \\ \dot{y}_3 = e \\ \text{let } y_4 = \dot{y}_3 = e \\ \dot{y}_4 = \dot{e} \end{cases}$$

$$\frac{u}{K_d} = \frac{K_p}{K_d} y_4 + \frac{K_i}{K_d} y_3 + \frac{K_d}{K_d} \dot{y}_4$$

$$\frac{de}{dt} = -\dot{y}_3$$

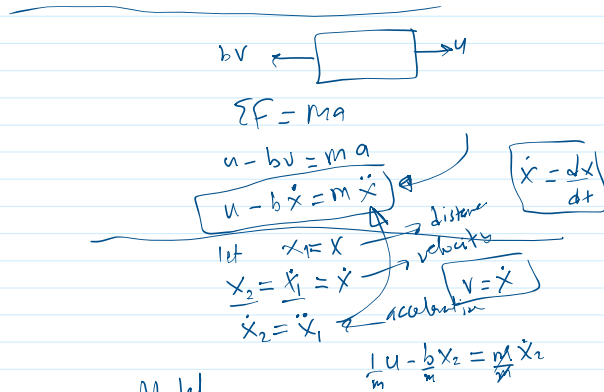
$$u = K_p e + K_i y_3 + K_d (-\dot{y}_3)$$

$$u = K_p e + K_i y_3 - K_d \left(\frac{1}{m}u - \frac{b}{m}y_2 \right)$$

$$u = k_p e + k_i y_3 + k_d (-\dot{y}_3)$$

$$u = k_p e + k_i y_3 - k_d \left[\frac{1}{m} u - \frac{b}{m} y_3 \right]$$

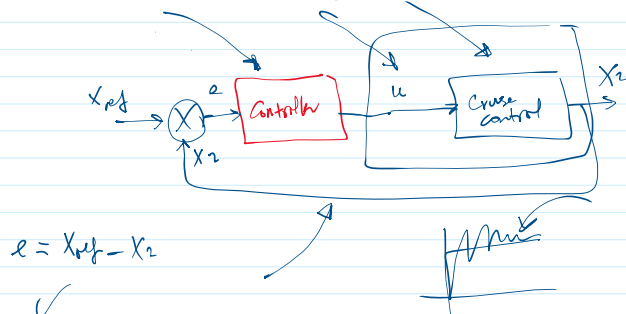
$$u \underbrace{\left(1 + \frac{k_d}{m}\right)}_{k_u} = \frac{k_p}{k_u} e + \frac{k_i}{k_u} y_3 + \frac{b}{m} y_3$$



Model

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -\frac{b}{m} x_2 + \frac{1}{m} u \end{aligned}$$

open loop



PI

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -\frac{b}{m} x_2 + \frac{1}{m} u \\ \dot{x}_3 &= e \\ u &= k_p e + k_i x_3 \end{aligned}$$

PI

$$e(s) \rightarrow \left[k_p + \frac{k_i}{s} \right] \rightarrow u(s)$$

$$u = \left(k_p + \frac{k_i}{s} \right) \times e$$

$$u = k_p e + \frac{k_i}{s} e$$

$$\rightarrow u(t) = k_p e(t) + k_i \int e(t) dt$$

let $x_3 = \int e(t) dt$
 $\dot{x}_3 = e(t)$